

Statistics - WS 27 P-Quiz

- **Quiz Type:** Practice Quiz
 - **Total Points:** 21
 - **Questions:** 21
 - **Due Date:** May 10, 2026
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Question 1 (1 pt)

Based on a recent poll of 1,015 U.S. households that have an Internet connection, the researchers claimed with 95% confidence that among all U.S. households that have an Internet connection, between 63% and 67% have a high-speed link. ☐ A. point estimation. ☒ B. interval estimation. ☐ C. hypothesis testing.

Question 2 (1 pt)

A company that provides coaching for the SAT claims in one of its ads: "90% of our students improve their SAT scores after attending our course." To check the company's claim, 500 students who took the company's course were sampled, and it was found that 407 of them (81.4%) actually improved their SAT scores after attending the course. Based on the sample results, we have some serious doubts regarding the accuracy of the company's claim. ☒ A. hypothesis testing. ☐ B. point estimation. ☐ C. interval estimation.

Question 3 (1 pt)

In a recent study, 1,115 males 25 to 35 years of age were randomly chosen and asked

about their exercise habits. Based on the study results, the researchers estimated that the mean time that a male 25 to 35 years of age spends exercising in a week is about 3.5 hours. ☐ A. interval estimation. ☒ B. point estimation. ☐ C. hypothesis testing.

Question 4 (1 pt)

A study on exercise habits used a random sample of 2,540 college students (1,220 females and 1,320 males). *The study found the following:* - 818 of the females in the sample exercise on a regular basis. - 924 of the males in the sample exercise on a regular basis. - The average time that the 1,742 students who exercise on a regular basis (818 + 924) spend exercising per week is 4.2 hours.

What is the point estimate for the proportion of all female college students who exercise on a regular basis? ($818 / 1220 \approx 0.6705$) ☐ A. 0.32 ☒ B. 0.67 ☐ C. 0.69 ☐ D. 0.48

Question 5 (1 pt)

(Based on the study above) Which of the following has a point estimate of 4.2? ☒ The mean time that all college students who exercise on a regular basis spend exercising per week. ☐ The mean time that all college students spend exercising per week. ☐ The percentage of all college students who exercise on a regular basis.

Question 6 (1 pt)

A researcher wanted to estimate μ , the mean number of hours that students at a large state university spend exercising per week. The researcher collects data from a sample of 150 students who leave the university gym following a workout. Which

of the following is true regarding $\bar{x}$? ☐ A. It is an unbiased estimate for μ . ☐ B. It is not an unbiased estimate for μ and probably underestimates μ . ☒ C. It is not an unbiased estimate for μ and probably overestimates μ .

Question 7 (1 pt)

Two opinion polls were conducted regarding the cultivation of Western ideals in academia. In particular, both polls were conducted in order to estimate p , the proportion of all Pakistani adults who believe that national educational institutions actively promote Western culture to the detriment of indigenous values. 1. *The poll by the Gallow company was based on a random sample of 1,024 Pakistani adults and estimated the proportion to be 0.60.* 2. *The New Research Center poll was based on a random sample of 2,002 Pakistani adults and estimated the proportion to be 0.57.*

Which of the two polls estimates p more accurately? ☒ A. The New Research Center poll. *(Larger sample size yields smaller standard error / higher precision).* ☐ B. The Gallow company poll. ☐ C. Both polls estimate p with the same accuracy because both are based on a random sample.

Question 8 (1 pt)

In a recent study, 1,115 males 25 to 35 years of age were randomly chosen and asked about their exercise habits. The researchers estimated the mean time that a male 25 to 35 years of age spends exercising with 90%, 95%, and 99% confidence intervals: (3, 4), (2.5, 4.5), and (2, 5). For which of the three intervals do you have the most confidence that it captures the population mean (μ)? ☐ A. (3, 4) ☐ B. (2.5, 4.5) ☒ C. (2, 5) *(The widest interval corresponds to highest confidence level, 99%).*

Question 9 (1 pt)

(Based on the question above) The confidence interval in which you have the most confidence that it captures the population mean μ must be: ☐ A. the 90% confidence interval. ☐ B. the 95% confidence interval. ☒ C. the 99% confidence interval.

Question 10 (1 pt)

(Based on the question above) Which of the three confidence intervals provides the most precise estimation? ☒ A. (3, 4) (*The narrowest interval*). ☐ B. (2, 5) ☐ C. (2.5, 4.5)

Question 11 (1 pt)

Suppose that based on a random sample, a 95% confidence interval for the mean hours slept (per day) among graduate students was found to be (6.5, 6.9). What is the margin of error of this confidence interval? (*Margin of error $m = \frac{6.9 - 6.5}{2} = 0.2$*) ☐ A. 6.7 ☐ B. 0.4 ☒ C. 0.2

Question 12 (1 pt)

A medical researcher wanted to estimate μ , the mean weight of newborns born to women over the age of 40. The researcher chose a random sample of 100 pregnant women who were over 40, and calculated the 95% confidence interval for μ to be (2935, 3135). If the researcher wanted to maintain the 95% level of confidence but report a confidence interval with a smaller margin of error, which of the following will achieve that? ☐ A. Redo the study, but this time with a sample of size 64. ☒ B. Redo the study, but this time with a sample of size 225. (*Increasing n reduces standard error $\sigma/\sqrt{n}$*). ☐ C. Redo the study and choose a different sample

of size 100.

Question 13 (1 pt)

In order to estimate μ , the mean weight of a typical passenger + luggage on an early morning flight from New York City to Washington DC, an airline company chose a random sample of 16 passengers. From past research, it is known that weights have a normal distribution with standard deviation $\sigma = 20$ pounds. Can we safely estimate μ with the formula for confidence interval that we developed? ☒ A. Yes (*Since the underlying population distribution is normal and σ is known*). ☐ B. No

Question 14 (1 pt)

For many years “working full-time” has meant 40 hours per week. Nowadays corporate employers expect more. To substantiate his claim, the researcher randomly selects 250 corporate employees and finds that they work an average of 47 hours per week with a standard deviation of 3.2 hours. In order to assess the evidence, we need to ask: ☐ A. how likely it is in a sample of 250 we will find that the mean number of hours per week corporate employees work is as high as 47. ☐ B. how likely it is that the true mean number of hours per week corporate employees work is 40. ☐ C. how likely it is that the true mean number of hours per week corporate employees work is more than 40. ☒ D. how likely it is that in a sample of 250 we will find that the mean number of hours per week corporate employees work is as high as 47 if the true mean is 40.

Question 15 (1 pt)

According to the Centers for Disease Control and Prevention, the proportion of U.S. adults age 25 or older who smoke is 0.22. A researcher suspects that the rate is lower among U.S. adults 25 or older who have a bachelor's degree or higher education level. What is the null hypothesis (H_0) in this case? ☐ A. The proportion of smokers among U.S. adults 25 or older who have a bachelor's degree or higher is less than 0.22. ☒ B. The proportion of smokers among U.S. adults 25 or older who have a bachelor's degree or higher is 0.22. ☐ C. The proportion of smokers among U.S. adults 25 or older who have a bachelor's degree or higher is not 0.22. ☐ D. There is no relationship between education level and smoking habits.

Question 16 (1 pt)

What is the alternative hypothesis (H_a) in this case? ☒ A. The proportion of smokers among U.S. adults 25 or older who have a bachelor's degree or higher is less than 0.22. ☐ B. The proportion of smokers among U.S. adults 25 or older who have a bachelor's degree or higher is 0.22. ☐ C. The proportion of smokers among U.S. adults 25 or older who have a bachelor's degree or higher is not 0.22. ☐ D. There is a relationship between level of education level and smoking habits.

Question 17 (1 pt)

The following two hypotheses are tested: - H_0 : The proportion of Pakistani adults who oppose coeducation is roughly 50%. - H_a : The proportion of Pakistani adults who oppose coeducation is above 50% (i.e., the majority oppose).

Suppose a survey was conducted with $n = 1,100$, and the p -value was found to be 0.002. The fact that the p -value = 0.002 means that: ☐ A. There is 0.002 probability of observing data like those observed. ☐ B. There is 0.002 probability that 50% of

Pakistani adults oppose coeducation. ☒ C. There is a probability of 0.002 (i.e., very unlikely) to observe data like those observed if the proportion of Pakistani adults who oppose coeducation were 50%. ☐ D. There is 0.998 probability that the majority of Pakistani adults oppose coeducation.

Question 18 (1 pt)

(Based on the question above) Based on the p -value ($0.002 < 0.05$), you can conclude that: ☐ A. the data provide significant evidence that the proportion of Pakistani adults who oppose coeducation is 50%. ☒ B. the data provide significant evidence that the majority of Pakistani adults oppose coeducation. ☐ C. the data do not provide enough evidence to conclude that the majority of Pakistani adults oppose coeducation. ☐ D. the data provide evidence that H_a is more likely than H_0 .

Question 19 (1 pt)

(Based on the question above) Say that the p -value was not given, but rather, the following conclusion was advertised: "The survey does not provide enough evidence to conclude that the majority of Pakistani adults oppose coeducation." Which of the following could have been the p -value that led to this conclusion? ☒ A. 0.125 (since $p > 0.05$, fails to reject H_0). ☐ B. 1.96. ☐ C. 0.045. ☐ D. -1.96 .

Question 20 (1 pt)

(Based on the question above) When would you conclude that the data provide enough evidence that the proportion of Pakistani adults who oppose coeducation is 50%? ☐ A. when the p -value is small (less than 0.05). ☐ B. when the p -value is not

small (above 0.05). ☐ C. when exactly half the individuals in the sample oppose coeducation and half support it. ☒ D. never. (*Hypothesis tests never accept/prove the null hypothesis as true; they only fail to reject it*).

Question 21 (1 pt)

In NBC News/Wall Street Journal poll conducted in 2010 determined that 61% of Americans did not support the Tea Party movement. In a poll of 1,000 Americans this year, 64% say they do not support the Tea Party movement. We tested $H_0: p = 0.61$ vs $H_a: p > 0.61$ at the 5% level. The p -value is 0.026, so we reject H_0 and conclude that opposition is greater than 61%. What type of error is possible here? ☒

A. Type I error (*Rejecting H_0 when H_0 is actually true*). ☐ B. Type II error. ☐ C. neither. ☐ D. both.